bizsim

User guide

Signing in, what each role sees, and what to click in every section: templates, companies, packages, people, sessions and the debrief.

Two jobs, two people: a BUILDER prepares templates, companies and packages; a FACILITATOR starts sessions from a published package and runs the debrief. An ADMIN does both.

A BUILDER cannot run a session; a FACILITATOR cannot edit what they run. The server enforces it, so the sections you can see are the sections your role owns.

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bizsim — user guide

Printable PDF: `docs/bizsim-user-guide.pdf` (regenerate with `python scripts/md_to_pdf.py docs/USER_GUIDE.md docs/bizsim-user-guide.pdf`, needs `fpdf2`).

How to sign in, which section each role sees, and what to click to get a class from "nothing prepared" to "debrief on screen".

The product has **two jobs done by two people**:

Job	Role	Does
Building	BUILDER	prepares templates (the economy), companies, people (the trainees), and packages; happens well before a class
Running	FACILITATOR	starts sessions from a published package, opens/computes rounds, runs the debrief
both	ADMIN	everything above
neither	PLAYER	the trainee: signs in, decides, submits

A `BUILDER` cannot run a session. A `FACILITATOR` does not build anything: the template, the companies, the people and the package all arrive prepared, and their job is only to run the class. A facilitator never seats anyone either: the class list is sealed inside the package, and a seat left open stays open. This is enforced by the server, not just hidden on screen — if you cannot see a section, it is the wrong account, not a bug.

1. Signing in

Interface language. The console ships in English and Chinese. Yours comes from, in order: the **class language of the package** you are seated under (trainees), your own choice on the account, the tenant default, then English. Template content — product names, trend labels, event text — is never translated (D23).

1. Open your site: `https://<your-domain>/` — the browser serves the console and forwards `/api` to the backend.
2. Fill the three boxes on the sign-in card:
   - **Tenant** — your organisation's code, e.g. `demo`
   - **Email** — the address your account was created with (case and stray spaces are forgiven)
   - **Password** — at least 8 characters; the first password is handed to you, change it later
3. Press **Sign in**.

Top left after signing in: a **language selector** (English / Chinese) and **sign out**. The language is saved on your account, so it survives a reload. It translates the interface only — the content you author (product names, event text, trend labels) is never translated.

Forgot the tenant code or have no account? An admin creates accounts (People → Add). If nobody can sign in at all, see §9 "First account" below.

2. What each role sees

Nav item	Builder	Facilitator	Admin
Packages	**yes**	**—**	**yes**
Templates	**yes**	**—**	**yes**
Companies	**yes**	**—**	**yes**
People	**yes**	**—**	**yes**
Sessions	**—**	**yes**	**yes**

3. Builder — Templates (the economy)

This is where a simulation is designed. There is one shipped template, `standard-v1` ("Standard manufacturing simulation"), marked **global**, and it is **read-only**.

3.1 Make your own copy

1. **Templates** in the left navigation.
2. Find the row **Standard manufacturing simulation** and press **Duplicate**.
3. A copy appears marked **yours** with status **draft** and *Based on* = the shipped key. It opens in the editor automatically; if not, click the row.

Everything below is edited **on your copy**. Press **Save** at the bottom of each tab. A banner confirms *"Saved. New simulations from this template use these values."*

3.2 The six tabs

Settings — the class-level numbers.

Group	Fields
pacing	rounds per simulation, minutes per round, cycles, rounds per cycle
starting position	starting cash, starting brand
costs	fixed cost per round, holding cost per unit, machine cost per worker
brand	brand decay per round
hiring	hiring cost (months), severance (months)
information	market review price, forecast horizon, forecast band
research	min R&D benefit ratio
dynamics	cycle demand drift, cycle cost drift

Group	Fields
quality (D40)	Offer quality upgrades to the class (checkbox, off by default), quality cost per point
Per function (table)	salary and starting headcount for Production, Sales, R&D, Admin

The quality switch is a per-template mechanic: **off means the trainee form has no quality card and any money sent to it is refused**. On, each company may buy quality points for a product at the price you set, and the level persists across rounds up to 100. A template written before this feature has no switch and therefore behaves exactly as it did.
Environment — the round-by-round timeline. One row per round: *demand index*, *cost index*, *wage index* (1.00 = normal), *trend strength*, *trend label* (free text, shown to trainees), *note*. Use **Add a row** / **Remove the last row** to change the length, then **Save** (saved whole).
Events — the event deck: surprises drawn once per run.

Column	Meaning
Code	unique short id, e.g. EV-FUEL-SPIKE
Name / Description	what the class sees
From round / To round	the window in which it can be drawn
Chance	probability 0.00–1.00 inside that window
Months	how long the effect lasts (1–12)
Demand ×, Cost ×, Wage ×	multipliers while it is active (1.00 = no change)
Target	one product code, or *all products*

Leave the deck empty for a calm economy. **Save** replaces the whole deck. The server refuses duplicate codes, a probability outside 0–1, `From` > `To`, and non-positive multipliers.
Products — one row per product: code, name, price, unit cost, quality, base demand, units per worker, income elasticity (how much a downturn hits this line), trend affinity.
Channels — code, name, reach weight, brand effect (how much brand helps in that channel).
R&D projects — code, name, target product (or all), advantage type (`UNIT_COST_DOWN`, `CAPACITY_UP`, `QUALITY_UP`, `ATTRACTIVENESS_UP`), size, true cost (hidden from trainees), duration in months, benefit curve (`LINEAR`, `POWER`, `CHANCE`), over-funding handling (`REFUND`, `LOST`).

3.5 Auto workers

Auto workers is a library of reusable opponents. Each has a **level** (Low / Mid / High — how well it plays) and a **personality** (Cautious / Balanced / Aggressive — what it wants). Creating one stores its name with the level as a suffix, so a scoreboard reads `Market 1_high`, and gives it an identity that can be seated like a person — but can never sign in.
To put one in a class:

1. Open the template you will publish from — **Templates → the template's name → Settings** — and tick
Offer auto workers to the class, then **Save**. It sits directly under the quality switch, at the end of the settings grid. Templates written before this switch existed keep playing without opponents until it is ticked; the global `standard-v1` is read-only, so duplicate it first if you need to change it.
2. In **Packages**, pick the template and the companies; the class list now offers the auto workers beside the trainees, labelled `· auto`.
3. Publish and start the session as usual. The opponent plays every round by itself, under the same submission rules, the same clock and the same validation as a trainee.

Note: the product, channel and R&D catalogue **locks once a session has used this template** — those codes are stored in every decision, so rewriting them would strand a live class. Settings, environment and events stay editable (each run freezes its own copy of them).

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4. Builder — Companies

A company profile is the reusable definition of a company: its **name** and **which products it carries**. Cash, brand and staff come from the template, not the profile.

1. **Companies** in the navigation.
2. The shipped profile `Standard manufacturer` is **read-only** — press **Duplicate** to get an editable copy, or press **Create profile** to make one from scratch.
3. Edit name, legal name and **product line** (`P1, P2` — or leave empty for the full line), press **Save**.

These profiles are what a package is built from, so make one per company that will compete.

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5. Builder — People (the class)

People are part of the preparation, exactly like the template and the companies (D30). The builder adds the trainees, because the class list travels inside the package and the facilitator only runs the session.

1. **People** in the left navigation.
2. **Add a trainee** — name, email, initial password (8+ characters). A builder only ever creates **Trainee** accounts: the Role selector is an admin's, and the server refuses staff roles for anyone else.
3. Hand the credential over; the trainee can change it after signing in.
4. Back in **Packages**, tick a company and pick the trainee in the class list — one person per company (D27). A seat left open means that company plays with nobody; nothing can fill it later.

- **Show trainees already seated in a company** — a switch above the list, **off** by default. With it off the list shows only the people still free to put in a company: anyone who already holds a seat in a package's class list or in a session is left out. Switch it on to see the whole roster again — that is what you need before resetting the password of someone who already played.
- **Set password** — the per-row button fixes a mistyped or forgotten credential (8+ characters).

A builder may reset a **trainee's** password; a staff account's password stays an admin's business.

- **Deactivate / Activate** — for someone who leaves, before or after playing. The login stops, the history stays readable.
- **Delete** — only an account that never played; once a decision was submitted the server answers *"deactivate the account instead"*.
- **Change a role** — admin only, in the same list.

A facilitator does not see **People** at all: the class arrives with the package they start.

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6. Builder — Packages (the hand-over)

A package is the immutable bundle you hand to the facilitator: **one template + the companies + the class list**.

1. **Packages** → **Publish a package**.
2. Give it a name used in class, e.g. *Autumn 2026*.
3. Choose the **Template** (your edited copy, not the global one).
4. **Class language** — English, Chinese, or *follow each trainee's own choice*. A trainee seated from this package reads the interface in that language for the whole session, whatever their own preference says. Their preference is not overwritten: the class language is scoped to the session and their stored choice is untouched.
5. Tick the **Companies** that will compete.
6. **Class list** — for each company, pick the trainee who will run it (accounts come from **People**). One person per company; the same person cannot sit in two competing companies. A seat left open means that company plays with nobody: **there is no way to fill it after publishing** (D33).
7. **Publish**.

The class list is part of the package because preparing happens before the class, not on the day. Publishing **freezes a private copy of the template** and records its version. Editing your draft afterwards changes nothing about this package — to change a published economy, edit the draft and publish a **new** package.

Retire stops new sessions from using a package; sessions already run from it keep their history. **Tidy up** appears when retired packages never ran a session; **Remove N** deletes them (and their frozen copies) after a confirmation. Packages that taught a class are never offered.

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7. Facilitator — running a class

7.1 Start a session

1. **Sessions** in the navigation. With no session selected you see **Start a session**.
2. Type a **Session name** (defaults to the package name).
3. Choose the **Package** — the line below shows the competing companies and the template version,

so you can confirm you are running what you think you are.

4. Check the **Class list** shown under it: one line per company with the person prepared for it (*seat empty* means that company has no trainee — nobody can fill it once the session starts).
5. **Start session** — everyone on the class list is seated automatically.

The session opens with one company per profile, its product line already set, and the prepared class already seated. If a package is missing, no builder has published one yet.

7.2 The run screen

Every company row on the scoreboard and in the debrief is labelled with the trainee seated in it (*Alpha Industries · Ana*), so "how did she do?" is answered without a separate roster — the seating itself stays with the builder and cannot be changed here (D33, D36).

Header: `Run <n>` · status · `round x of y` · cycles. There is **one** round button, because only one of its two jobs is ever the right move (D38):

What the button says	What it does
Open round N	nothing is open: opens round N for decisions (round 0 → 1 → …)
Compute round	a round is open: closes it and calculates results for every company
Session finished (greyed)	the last round is computed; nothing more to open

One round is therefore two clicks with a pause in between — **Compute round**, read the debrief, then **Open round N+1**. **Delete run** removes the session and everything played in it — **accounts survive**.

You can only have **one session in progress at a time**. To run two cohorts in parallel you need two facilitator accounts.

7.2.1 The round clock, the submissions and the audit trail

Under the header there is a **Round clock** card. It is the clock that closes the round automatically when the time you set in the template runs out (D12):

Button	What it does
Pause clock	stops time. The remaining time stops counting, nothing computes, and trainees can still submit. Use it for a discussion in the middle of a round.
Resume clock	starts it again and gives back exactly the time it was stopped — a clock paused for six minutes runs six minutes longer.
Add 5 minutes	moves the deadline out by five minutes (works while paused, and on a round that had no clock).

Void round N appears when nothing is open and at least one round has been computed. It undoes the last computed round: every company goes back to the state it had before that month, that round's results and statements are deleted, and the round opens again so the class can correct the decision and you can compute it again. The decisions themselves are kept — the class corrects a

month, it does not retype it. You are asked for a reason, which is kept in the audit trail. Rounds computed before this feature existed carry no snapshot and are refused with *"cannot be restored"* rather than half-undone.

Submissions lists every company with *submitted* or *not yet*, who sent it, and **show the month** expands exactly what they asked for (volumes and prices, marketing, R&D, quality, staff, machines).

Adjust cash on the same row moves one company's balance by hand: type the amount (**negative to fine, positive to lend or compensate**) and a reason of at least three characters, then **Apply**. The change is immediate and lands in the audit trail; the class can see it.

Audit trail is the session's log of every hand laid on it — clock paused/resumed/extended, cash adjusted, round voided — with the round, the company, the amount, the reason, who did it and when. When a class asks "why did our cash jump?", this table is the answer.

7.2.2 The opponents

When a session has auto-played companies, **Auto workers this session** lists one row per opponent per round: the brain and its version, how it was called (**scheduled**, **by hand**, or **missed**), and its one-line reason. **Replay** re-runs the same input and compares — *"the same input produces the same decision"* means the trace can be trusted in front of the class; a difference means the brain was edited after that round.

Ask them to move now makes the opponents submit immediately instead of waiting for their moment. It is recorded as *by hand*, so the log stays honest.

7.2.3 The class table

Class standings on the run page is the exact table the class reads — rank, company, profit — and the switch that decides whether they may read it. It starts **hidden**: press **Show the class** after reading the debrief, and every computed round is then visible to the trainees; **Hide from the class** takes it away again. Both moments are in the audit trail.

7.3 Scoreboard

One row per company for the latest computed round: profit, cash, share, stock. Empty until you **Co mpute round**.

7.4 Debrief

Under the scoreboard, for any computed round:

1. Pick the **Round** in the selector.
2. Read across: revenue, cost, profit, cash, share, lost sales, capacity used/available.
3. **Detail** on a company expands its month, in reading order:

- **Submission · round N** — what the company asked for: units, price and quality spend per product, marketing per channel, R&D per project, production staff and machines, and whether the market review was bought. A company that sent nothing says so.
- **Results · round N** — then a three-column breakdown, so the arithmetic reads left to right:
- **Revenue** — revenue for the round, and units lost to stock-outs

- **Cost** — cost of goods sold, payroll, marketing, R&D, fixed cost, stock holding, hiring/firing, capacity, market review, then **Total cost**
- **Result** — opening cash (the balance the round started from), profit, any stock and capacity movement, then cash after the round, brand after the round, capacity used, market share

Opening cash + profit + stock/capacity movement = cash after the round, and each round opens exactly where the previous one closed — that is the continuity line. The same breakdown is shown to the trainee for their own company (§8), so a learner can go over their round outside class. Revenue − Total cost = Profit, on screen, with no line hidden: anything the engine charged that is not one of the named lines appears as **Other charges** instead of being folded into a total.

Under the three columns, **Per product** breaks the same round down line by line — one row per product with **Opening**, **Produced**, **Demand**, **Sold**, **Lost**, **Closing**, **Revenue** and **COGS**. This is the answer to "82 pieces were lost — of which product?" and "0.94 left in stock — of which product?": a company that stocked out on one product and kept a little of another shows both, separately. The rows add up to the totals above, because they are the same stored lines, not a second calculation. Units are whole pieces (D34); the market share stays a percentage of the whole market.

4. The **Class** footer sums revenue and profit for the whole class.

Every number comes from that round's statements — the debrief cannot disagree with what the engine charged.

7.5 Presenting and exporting the debrief

Three buttons sit next to the round selector:

Button	What you get
Present	a full-screen classroom view of the selected round. Page 1 is the six numbers you read out — profit, cumulative profit, market share, cash, brand, stock — one card per company, labelled with the trainee. Page 2 is the explanation: lost sales, capacity used, R&D completed, status. ← → change page, ↑ ↓ change round, Esc closes.
Download PDF	an A4 landscape report of the whole term. It opens with two summaries — *Summary by trainee* (one page per company: its rounds, then a Total line) and *Comparison by month* (one page per round, every company side by side) — and then gives the detail behind each month: what each company sent and every cost line. Company and trainee names are rendered as authored, including Chinese.
Download Excel	the same numbers as a workbook: By trainee (one row per company across the term: rounds, totals, final cash, average share), Results (one row per company per round), Submissions (the plan per company per round) and Cost lines, each with a Round column so the whole term can be filtered or pivoted.

Both files are built from the stored results, never a recomputation, so a downloaded report matches what the class saw on screen. **The document is written in the language you are using** — switch the console to Chinese and the download comes out in Chinese, labels and sheet names included. Company, product and trainee names stay exactly as they were entered. Present view is

also what prints cleanly: Ctrl/Cmd+P in **Present** mode hides the console and keeps the cards.

7.6 Live

A feed of what has happened in the session (`round.opened`, `decision.submitted`, `round.closed`).

8. Trainee — playing a round

1. Sign in with the trainee account (tenant + email + password).
2. The decision screen shows the company, **Round x of y**, cash, brand, machines — but only while a
round is **open**. Before the facilitator opens round 1 you get a waiting card (*"Waiting for your facilitator to open round 1"*) with **no form and no submit button**; the same happens between rounds. The page re-checks every 10 seconds, so nothing needs reloading once they open it, and your results from rounds already computed stay readable underneath.
If you see **"No simulation yet"**, you are not on any class list: the builder seats you in the package's class list and publishes a new package; the facilitator cannot seat you during a session.
The **round clock** is on your screen while a round is open: *12:34 left in this round*, turning red under a minute, and *Paused at 12:34* while your facilitator has stopped time. It counts down on its own — you do not need to reload.

What it means	Who fixes it
you are not seated in any session	builder: seat you in the package class list (People → then Packages)
you are seated but no round is open yet	you see the waiting card; facilitator: Open next round
the round was computed and the next one is not open	you see the waiting card; facilitator: Open next round

3. Fill in:
   - **Production and price** — units to produce and the price per product (stock and unit cost are shown for reference)
   - **Workforce and capacity** — production staff
   - **Marketing** — spend per channel
   - **Research and development** — investment per project, *this month*. A project that has already finished (**COMPLETED** or **FAILED**) shows a closed field and *"No further investment"*: the money cannot be spent again, and leaving it at 0 does not block the round.
   - **Quality upgrade** — only when your template offers it (D40). One line per product: the level the product carries now, how much you spend this month, and the points that money buys. The level stays for every later round and stops at 100; money beyond that is not charged. The spend is a cost of the month, like R&D, so it lowers profit and cash by the same amount.
   - **Market review** — optional paid report; the cost is deducted from the result
4. **Submit round n**. Submitting again replaces the decision until the round is computed.

5. **Your results** — once a round is computed, the trainee gets the same summary and detail the class debrief shows for their company:

- a **summary table**, one row per computed round (revenue, cost, profit, cash, lost sales, capacity used) — click a row to choose it;
- the chosen round in **detail** under Revenue · Cost · Result;
- **Per product**, the same round broken down by product (opening, produced, demand, sold, lost, closing, revenue, COGS) — this is where "which product lost those pieces?" is answered for the trainee too.

Competitors' figures and the market share stay on the facilitator's screen; the trainee sees only their own company. Everything comes from the same statement rows the debrief reads.

6. **Class standings** — above your results, once your facilitator has shown it and a round has been computed: rank, company and that round's profit for every company in the class, with your own line highlighted. That is all a trainee sees of anyone else — no market share, no cash, no statements. Your facilitator decides when it appears.

7. **My results (PDF / Excel)** in the same card gives the trainee their own report, **one page per round**: that month's summary, then what they sent, then the cost lines — their company only, no market share, no competitor line. A trainee who has not been seated in the session gets a refusal, not an empty file.

Errors come back in your own language and name the field, e.g. *"P1: the price must be greater than zero"* or *"Not enough cash: the plan needs X, you have Y."*

9. Admin — accounts

People lists every account in the tenant with role and status. Building and admin see it; a facilitator does not (D30).

- **Add a trainee** — name, email, initial password (8+), and the **Role** selector: **Trainee** / **Facilitator** / **Builder** / **Admin**. Only an admin sees the role selector; the server refuses staff roles for anyone else.
- **Set password** — a per-row button. Staff may reset a **trainee's** password; only an admin may reset a staff password. Use it when someone cannot sign in: a mistyped credential at creation, a forgotten password, or a password handed to the wrong person. The new password needs 8+ characters and is stored hashed; hand it over and they can change it again after signing in.
- **Change a role** — the role cell in the list is a dropdown for admins. Pick the new role and it saves immediately; the account keeps its password and its history. Two guards: you cannot change your **own** role (ask another admin), and the **last active admin** cannot be demoted — create a second admin first.
- **Deactivate** stops a login and keeps their history — use this when a person leaves. **Activate** reverses it.
- **Delete** removes an account that never played. Once someone has a computed decision, deletion is refused with *"deactivate the account instead"* — that is deliberate, history must stay truthful.

First account (fresh deployment)

The server creates the first account from three environment variables on start:

```
BIZSIM_BOOTSTRAP_TENANT       e.g. demo
BIZSIM_BOOTSTRAP_EMAIL        your email
BIZSIM_BOOTSTRAP_ADMIN_SECRET 8+ characters
BIZSIM_BOOTSTRAP_ROLE         optional, defaults to ADMIN
```

Keep it an **ADMIN**: only an admin can create the builder and facilitator accounts. Existing accounts are never rewritten by this; remove the variables once you have signed in.

10. Rules that surprise people

Rule	Why
The shipped template is read-only	every class would otherwise be playing a different economy
Duplicate to edit	your copy is yours; the base stays a reference
Catalogue locks after a session used the template	decisions store product codes
Published packages are frozen	a class that was prepared cannot be changed mid-term
Re-publishing makes a new package	the old one stays as it taught
One session per facilitator	a second parallel cohort needs a second account
Sessions are disposable, accounts are permanent	deleting a run keeps the people
A builder cannot see sessions	preparing is not watching

11. If something looks wrong

What you see	Cause / fix
Unauthorized / invalid or expired token	sign out and in again; the token lasts 8 hours

What you see	Cause / fix
sign-in says the password is wrong	the tenant code and email are now matched ignoring case, so check instead: (1) the account is active — a deactivated account gives the same answer, (2) the password actually has 8+ characters, (3) no stray spaces pasted in. Admin fix: People → Set password
This action is for facilitators	you are signed in as a builder; use a facilitator account for sessions
only a builder prepares…	the account is a facilitator; templates/companies/packages need a builder or admin
No published packages yet	a builder has not published one; see §6
you already have a simulation in progress	finish or delete the current run, or use another facilitator
the shipped profile is read-only	duplicate it first
no Role dropdown when adding someone	you are signed in as a facilitator, or the deployed build predates the selector — an admin sees it; otherwise redeploy (§11, api/version)
after a redeploy the screens look unchanged	the console's index.html was cached. Hard-refresh (Ctrl+Shift+R) or use a private window; from this release it is served no-store and cannot stick

What you see	Cause / fix
no admin account at all (everyone is a facilitator or trainee)	set BIZSIM_BOOTSTRAP_EMAIL=<existing account> + BIZSIM_BOOTSTRAP_PROMOTE=true in Dokploy, restart, sign in, remove the variables — role only, password unchanged
created an account with the wrong role	People → the role cell is a dropdown for admins → pick the right role; no need to delete and recreate
this package has been retired	publish a new package
Blank screen, or a feature you were told exists is missing	check https://<domain>/api/version: builtAt is the build time of the running image. If it predates the change, redeploy. latest only tracks database migrations, so a UI-only release does not move it

12. Run sheet for a class

Before (builder, days earlier)

1. Duplicate the template, adjust settings / environment / events / catalogue, **Save** each tab.
2. Create or copy the company profiles.
3. Create the trainee accounts (People) and a facilitator account — an admin does this.
4. Publish a package with those companies **and the class list** (one trainee per company).
5. Hand the package name to the facilitator.

On the day (facilitator)

5. Sign in → **Sessions** → **Start a session** → pick the package → tick trainees → **Start session**.
6. **Open next round** → trainees decide and submit → **Compute round**.
7. Repeat for each round; watch the **Live** feed.
8. **Debrief** → pick the round → walk the class through revenue, cost and profit, expand **Detail**.

After

9. **Delete run** if you want a clean slate (accounts stay), or leave it as history.
10. **Tidy up** on Packages removes retired, never-used packages.